

Introduction to Einstein's General Relativity

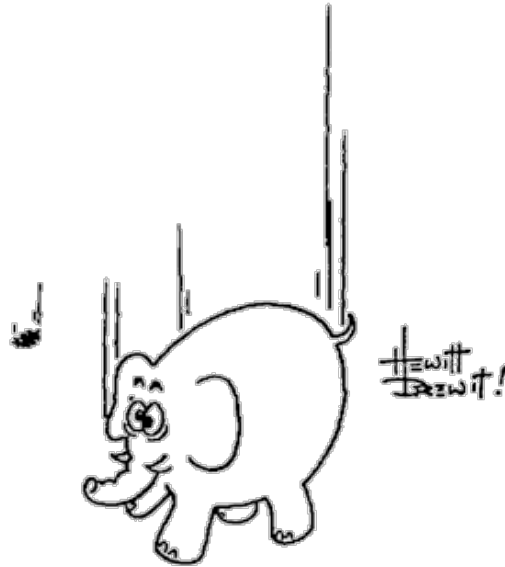
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Abstract

These are Lecture Notes for PHYS 413/513 at Rice University. At the moment, they constitute a work in progress. Please report any typos/errors to mustafa.a.amin@rice.edu.

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Preface

These are notes for a one semester course, PHYS 413/513, at Rice University. The course is meant for students (undergraduate and graduate) who have little or no prior exposure to Einstein's General Relativity. However, the students are expected to have a solid foundation in classical mechanics (with Lagrangians), multivariable calculus, differential equations, and linear algebra. Familiarity with Special Relativity and Electrodynamics will be very helpful too.

Students: Do the recommended reading before class!

Acknowledgements

The primary textbook for the course is James Hartle's "Gravity". The notes rely partly on the textbook for organization, and are very much a work in progress. Don't expect complete notes at the moment. I have added my own examples, explanations and visualizations. where I thought I had something to say which can help students further. I will also supplement these notes with exercises to be done as you read through. This will all happen in due time.

I learnt General Relativity first by reading Sean Carroll's more mathematical textbook, which I thoroughly enjoyed. I was enamoured by the elegant differential geometry at the time, but missed some of the relevant physics in that first pass. Hartle's book helped address that for me. I have also benefited from Misner, Thorne and Wheeler's "Gravitation", Weinberg's presentation in "Gravitation and Cosmology" and Wald's book "General Relativity". More recently, I have enjoyed learning from the Special and General Relativity sections of Thorne and Blandford's "Modern Classical Physics".

I owe an immense gratitude to my teachers – Prof. Eva Silverstein who first asked me to learn GR during a "rotation" during the first year of graduate school, Prof. Renata Kallosh for my first course, and Prof. Robert Wagoner for letting me TA for his GR class. This is how I learnt GR during graduate school. Prof. Barbara Shipman at the University of Texas at Arlington taught me multivariable Calculus and Linear Algebra, and then differential Geometry during my undergraduate years, both in class and often on Friday afternoons beyond class. Her enthusiasm for and pedagogy was infectious, and I hope I can follow in her footsteps.

And, importantly, the previous Phys 413/513 students at Rice were critical in making the course what it is now.

In preparing these notes, I am using some LaTeX and editing assistance from ChatGPT.

CONTENTS

1	Invitation	4
1.1	Going along with Gravity	4
1.2	The road ahead	6
1.3	Newton vs. Einstein	6
1.4	What can GR do for you?	8
1.5	Cube of Physics	9
2	Geometry as Physics	10
2.1	Euclidean vs. intrinsically curved spatial geometry	10
2.1.1	Two dimensions	10
2.1.2	Three dimensions	11
2.2	Specifying a geometry	12
2.2.1	Labeling and calculating with co-ordinates	12
2.3	Spatial metric	13
2.3.1	Path length and volumes	14
2.4	Practice Problems and Beyond	16
2.5	Appendix	17
2.5.1	Curves of Extremal Length	17
3	From Newtonian Space and Time to Minkowski Spacetime	19
3.1	Newtonian Space, Time and Reference Frames	19
3.1.1	Inertial Reference Frames	19
3.2	Galilean Transformations	20
3.3	Principle of Relativity	20
3.3.1	“Light” Trouble for Galilean Transformations	21
3.3.2	“Light” Trouble for Galilean Transformations	21
3.4	Resolution via Lorentz Transformations	21
3.5	The “Price”	22

3.6	Invariant spacetime interval and Minkowski spacetime	23
3.7	Moving in Spacetime	23
3.8	Proper Time	24
3.9	Spacetime Diagrams	24
3.10	Newtonian Physics as the Low-Speed Limit	25
3.11	Take Away	26

CHAPTER 1

INVITATION

We will explore one of the most impressive achievements of our species. Along the way, I will also tell you how to lose some weight.

1.1 Going along with Gravity

Launch a coconut and a tomato with the same velocity from the earth's surface. In absence of air resistance, they will follow the same trajectory!¹ This, in itself is surprising – the trajectory is independent of their mass. If you were launched with them with the same velocity, they would simply appear to float next to you – gravity seems to have disappeared! More generally, if you were falling freely under gravity, by looking at the tomato and coconut (for a short enough period of time) they would each appear to be moving at a constant velocity, with no gravitational acceleration. Even if you, the coconut and tomato were launched with different individual initial velocities, you would still find that they appear to move at a constant velocity relative to you and each other. You will also “feel” weightless.

Exercise 1.1.1 Why is it surprising that in the gravitational field $\mathbf{g}$ the coconut and tomato, which have different masses m_c and m_t , follow the same trajectory? Here $|\mathbf{g}| = |-\nabla\Phi| \approx GM_e/R_e^2$. Consider the situation where the (approximately) constant gravitational field $\mathbf{g}$ near the earth surface is replaced by a constant electric field $\mathbf{E}$, and assume that the fruits have charges q_c and q_t . The accelerations would be $\mathbf{a}_c \propto q_c/m_c$, and similarly $\mathbf{a}_t \propto q_t/m_t$. It depends on the ratio q_i/m_i , ie. on the nature of the bodies, so the trajectories will be different! However, in the case of gravity, $\mathbf{a}_c = \mathbf{a}_t = \mathbf{g}$ independent of the mass! Check these statements using Newton's second law, as well as gravitational and electrostatic forces. See Fig. 1.1

¹Check out a version of this experiment on the moon <https://www.youtube.com/watch?v=KDp1tiUsZw8>. Also see <https://www.youtube.com/watch?v=E43-CfukEgs>.

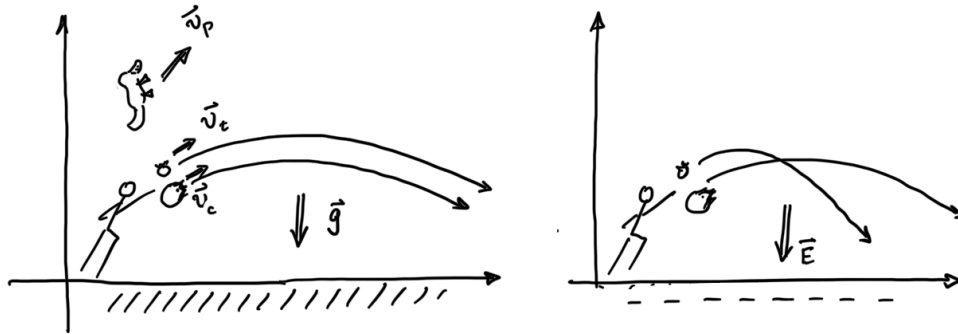


Figure 1.1

Exercise 1.1.2 Suppose that the fruit were launched from the surface of the earth with velocity $\mathbf{v}_c$ and $\mathbf{v}_t$. They will follow parabolic trajectories $\mathbf{x}_c(t)$ and $\mathbf{x}_t(t)$, with velocity $\mathbf{v}_c(t)$ and $\mathbf{v}_t(t)$ and acceleration $\mathbf{a}_c(t) = \mathbf{a}_t(t) = \mathbf{g}$. (i) Write down $\mathbf{x}_{c,t}(t)$ and $\mathbf{v}_{c,t}(t)$. Now imagine at the same time, a platypus named Polly starts falling freely from some height h , (ii) write down $\mathbf{x}_P(t)$, $\mathbf{v}_P(t)$ and $\mathbf{a}_P(t)$. (iii) What would the position, velocity and acceleration of the coconut and tomato be as measured in the “falling frame” of Polly? You should find that $\mathbf{g}$ disappears; Polly will conclude there is no gravity in the situation. (iv) Let the initial velocities $\mathbf{v}_c = \mathbf{v}_P = 0$. The coconut and Polly are falling from a height $z_c(t=0) = h_c$ and $z_P(t=0) = h_P$, both starting at rest. In this case make a sketch of t vs. $z_c(t)$ in the earth’s rest frame. Then make the same sketch in Polly’s rest frame. What do you notice? Repeat the same exercise with non-zero, and different initial velocities for Polly.

While not obvious to us on earth easily (unless you are the free, but unfortunate Polly), get in touch with your favorite “freely falling” astronaut in the international space station to understand how gravitational effects (mostly) disappear for them in their “local” laboratory on the space-station.²

²In 2024, I read a sign at the Johnson Space Center (inside the mock shuttle), which said astronauts float because of lack of earth’s gravity since they are far away from the earth. After this course, you should tell me what you think about this statement.

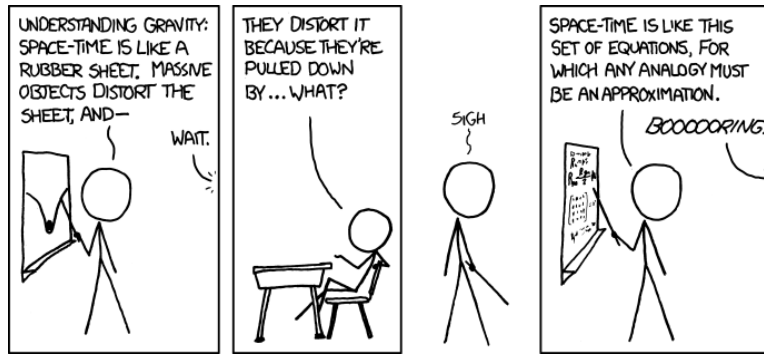


Figure 1.2: Credit: <https://xkcd.com>

With Einstein as our guide, these simple observations lead us to an astonishing insight into Nature. Gravity is unlike other known forces – it is really about the nature of space and time. Three dimensional space is not a passive arena in which events take place in time, but space and time together – *spacetime* – is an active, dynamical entity. It is a smooth collection of events: a 3+1 dimensional spacetime, whose geometry is determined by the mass-energy within it. The mass of the earth “curves” the spacetime around it.

As we will learn, despite the parabolic locus of the trajectory in space, the coconut and tomato are moving in this curved *spacetime* on the “straightest” possible path which is independent of the composition and mass of the fruit. Gravity is a manifestation of this curved spacetime.

1.2 The road ahead

We will spend much of the course developing a quantitative, calculable and testable understanding of the above ideas. Our underlying theoretical framework for gravity is that of Einstein’s General Theory of Relativity (GR). This theory, arguably one of humanity’s most elegant and well tested ideas, is built upon Einstein’s Special Theory of Relativity. We will review Einstein’s Special Theory of Relativity (SR) first – with an emphasis on geometry. The path to GR then is a matter of living with SR “locally” in small enough regions of spacetime, and developing an understanding of how to connect the happenings in many such small regions together. We will compare and contrast our developing understanding of GR with Newtonian ideas of gravity, and see why Newtonian ideas fail to be consistent with SR and more importantly, with observational and experimental tests. Nevertheless, it will be helpful to keep the following connection between Newtonian and Einstein gravity at the back of our mind: $\ddot{\mathbf{x}} = -\nabla\Phi \rightarrow$ geodesic equation, $\nabla^2\Phi = 4\pi G\rho \rightarrow$ Einstein Equations.

1.3 Newton vs. Einstein

The Newtonian theory of gravity is of course still extremely useful and accurate for most applications on the earth, solar system and in astrophysics more generally. However, Newtonian gravity fails when considering systems with mass M and extent R , such that GM/Rc^2 becomes order unity. Here $c = 3 \times 10^{10} \text{cm s}^{-1}$ is the speed of light in vacuum and $G = 6.67 \times 10^{-8} \text{dyne gm}^{-2} \text{cm}^2$

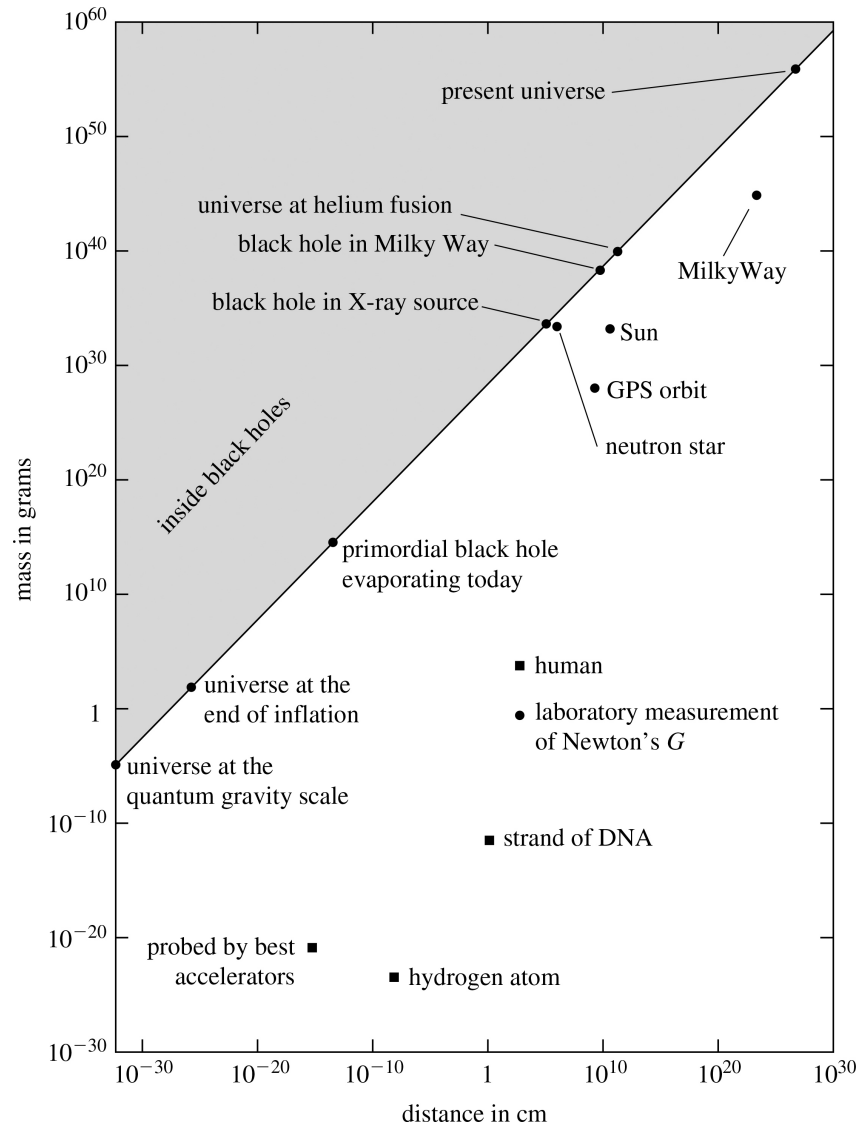


Figure 1.3: General relativity effects are not ignorable near the diagonal line, where $2GM/Rc^2 = 1$. Figure from J. Hartle. To understand the diagonal line, recall that the escape velocity from the surface of a body of mass M and radius R is $v_{\text{esc}} = \sqrt{2GM/R}$. When M and R are such that this escape velocity $v_{\text{esc}} \rightarrow c$, even light cannot escape from its surface (this is the line $M = (c^2/2G)R$). This is the historical definition of a “black hole” (Michell & Laplace, 18th century).

is Newton’s gravitational constant. Near a Black Hole, $GM/Rc^2 \sim 1$, neutron star $\sim 10^{-1}$, near the sun (say at the orbital radius of mercury) it is $\sim 10^{-8}$ and $\sim 10^{-6}$ near the sun’s surface, whereas near the earth’s surface $GM/Rc^2 \sim 10^{-9}$. Note that near the earth and the sun, the precision of the measurements is sufficiently high that even for such small values of GM/Rc^2 , we need to take deviations from Newtonian gravity into account. See Fig. 1.3 to get a feel for where different objects lie in the size-mass plane, and where how far they are from the $GM/Rc^2 \sim 1$

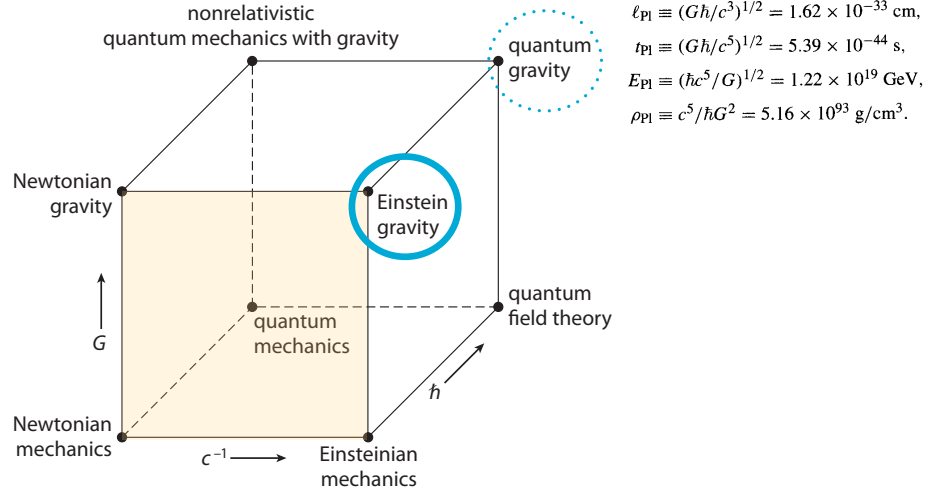


Figure 1.4: Cube of Physics, adapted from A. Zee.

region.

It is very useful to remember that for the sun $GM_\odot/c^2 \sim 1 \text{ km}$, where $M_\odot \sim 2 \times 10^{30} \text{ kg}$.

Exercise 1.3.1 Estimate $GM/(Rc^2)$ for (i) a proton (ii) human (iii) moon (iv) earth (v) sun (vi) neutron star.

1.4 What can GR do for you?

In the course, we will learn to appreciate (aspects of) the working of the Global Positioning System (GPS) relevant for your smart phones to help you orient yourself. Moving further out from our abode on our remarkable third rock from the sun, we will learn how the geometric understanding of spacetime explains the puzzling orbit of Mercury around the sun, and the observed deflection of light by the sun.³ It also describes why we can occasionally see the same supernova go off multiple times – seeing the “echoes” of the same explosion. We will be able to describe the formation and environment of spectacular ultra-compact “objects”, neutron stars and black holes in our cosmos. We will be able to understand how a dance of their binaries generate ripples in our dynamical spacetime, which have now been detected on our rock.⁴ Time-permitting we will also touch upon how the evolution of the universe as a whole relies on our understanding of the laws governing a dynamical spacetime.

³Light deflection from the sun was first observed during a total solar eclipse of 1919. For a bit of history and its significance, you might like a talk I gave on this in the context of the solar eclipse that was viewable from the continental US in 2024. <https://www.youtube.com/watch?v=D1LGSSoLfNg>.

⁴See <https://www.nobelprize.org/prizes/physics/2017/press-release/>.

1.5 Cube of Physics

To conclude, let us also put general relativity in the context of classical and quantum mechanics. A nice visual way is via the “cube of physics”, see Fig. 1.4. You should read this as, for example, when G becomes more relevant, then Newtonian gravity plays a prominent role. We will not have much to say about when quantum gravity in this course. It becomes important at the Big Bang and near the center of black holes. Notice the physical scale where quantum gravitational effects need to be taken into account. For this course, we will restrict ourselves to classical gravity.

GEOMETRY AS PHYSICS

Gravity and Geometry Considerations of universal free-fall, amongst other things, eventually led Einstein to the profound insight that gravity is not a force; it is a manifestation of the *geometry of spacetime*. He was helped by the fact that his Special Theory of Relativity (ignoring gravitational effects), initially formulated in terms of transformation laws between different observers, had already been geometrized by Minkowski.¹ This geometry, whether in Special or General Relativity, is $3 + 1$ dimensional (3 spatial and 1 temporal dimension). In General Relativity, spacetime can be intrinsically curved, and this curvature encodes gravitational effects.

Geometry of space and spacetime is something accessible to experimental verification. Gauss could in principle have tested whether the angles of a large triangle sum to π (the triangle is in three dimensions, not restricted to be on the earth’s surface).² Modern cosmological observations perform the same kind of experiment on vastly larger scales by comparing known physical scales with their observed angular sizes. You can learn about Gauss’s failed attempt to do so in the vicinity of earth. And then learn about our very successful attempt to do so in the context of cosmology. Both are described in Chapter 2 of Hartle.

2.1 Euclidean vs. intrinsically curved spatial geometry

To understand “intrinsically curved” geometry, let us first restrict ourselves to spatial geometries (as opposed to spacetime geometry). To make things easier still, let us consider 2 dimensional spatial geometries.

2.1.1 Two dimensions

2d Euclidean An example of a Euclidean geometry is a 2-dimensional plane, where Euclid’s axioms are satisfied. For example, (i) parallel lines do not intersect, (ii) sums of interior angles of

¹Einstein initially resisted Minkowski’s geometrization of Special Relativity, reportedly dismissing it as “superfluous learnedness.” He is also quoted as saying, “Since the mathematicians have invaded the theory of relativity, I do not understand it myself anymore.” The geometric viewpoint later became indispensable to Einstein in developing general relativity.

²This is historically debated a bit.

a triangle made of straight lines is π radians and (iii) the ratio of the circumference of a circle to its radius is 2π . Non-Euclidean geometry is of course a geometry where any of Euclid's axioms are violated.

2d intrinsically curved An example of an intrinsically curved 2-dimensional geometry is a 2-sphere (with radius a). Note that Euclid's axioms are not satisfied on a 2-sphere. For example: (i) Any two distinct great circles intersect, so there is no global analogue of distinct parallel straight lines on the sphere. (ii) Sums of interior angles of a triangle made of straight lines (great circles) is $> \pi$ radians ($= \pi + A/a^2$). Here, A is the area of the triangle on the 2-sphere. (iii) The ratio of the circumference of a circle to its radius is $< 2\pi$. That is, $\text{circumference}(C)/\text{radius}(r) = 2\pi \sin(r/a)/(r/a)$ where r is the radius of the circle on the 2-sphere.

Note that these properties can be tested by a bug living on the surface of the sphere, without ever leaving the surface. These are “intrinsic” properties of the surface, not requiring any information beyond the surface itself. For example, a bug could take a small segment of a string, and attach one end at some point on the sphere. Then crawl away till the string becomes taut. The bug would measure this taut length in terms of their own tiny steps. This is a radius r in units of their step size. Now, the bug can walk around keeping the string taut to trace out a circle on the sphere. Again, the circumference C can be measured in terms of the number of their tiny steps. By comparing the circumference to the radius, the bug would discover that they were living in a non-Euclidean space. They never had to see the embedded 2-sphere in 3d Euclidean space by flying out of the surface! Also note they can look for these effects without knowledge of global properties such as whether their space is finite or infinite.

A geometry is *intrinsically curved* if departures from Euclidean geometry can be detected by measurements made entirely within the space, even locally, without reference to an embedding in a higher-dimensional space. The opposite is *intrinsically flat*. Thus, a sphere is intrinsically curved, while a plane is intrinsically flat. A cylinder provides an instructive example: although it looks curved when embedded in three dimensions, it is intrinsically flat.

Intrinsically curved spatial geometries are necessarily non-Euclidean, although the term “non-Euclidean” by itself can also include global differences that do not involve intrinsic curvature.³

2.1.2 Three dimensions

3d Euclidean Our usual Euclidean 3d space is the obvious generalization of a plane from 2 to 3d.

3d intrinsically curved A 3-sphere is a 3d generalization of the 2-sphere. It is also non-Euclidean. For example the volume enclosed by a 2-sphere of radius r in this 3-sphere geometry is

$$V = 2\pi a^3 \left[\frac{r}{a} - \sin\left(\frac{r}{a}\right) \cos\left(\frac{r}{a}\right) \right]. \quad (2.1.1)$$

In the limit $r/a \ll 1$, this volume approaches its Euclidean counterpart $V = (4\pi r^3/3)[1 - (1/5)r^2/a^2 + \dots]$.

³Euclidean geometries are intrinsically flat, but the converse need not hold globally. For example, an infinite cylinder is intrinsically flat and locally Euclidean, but is not globally equivalent to the Euclidean plane.

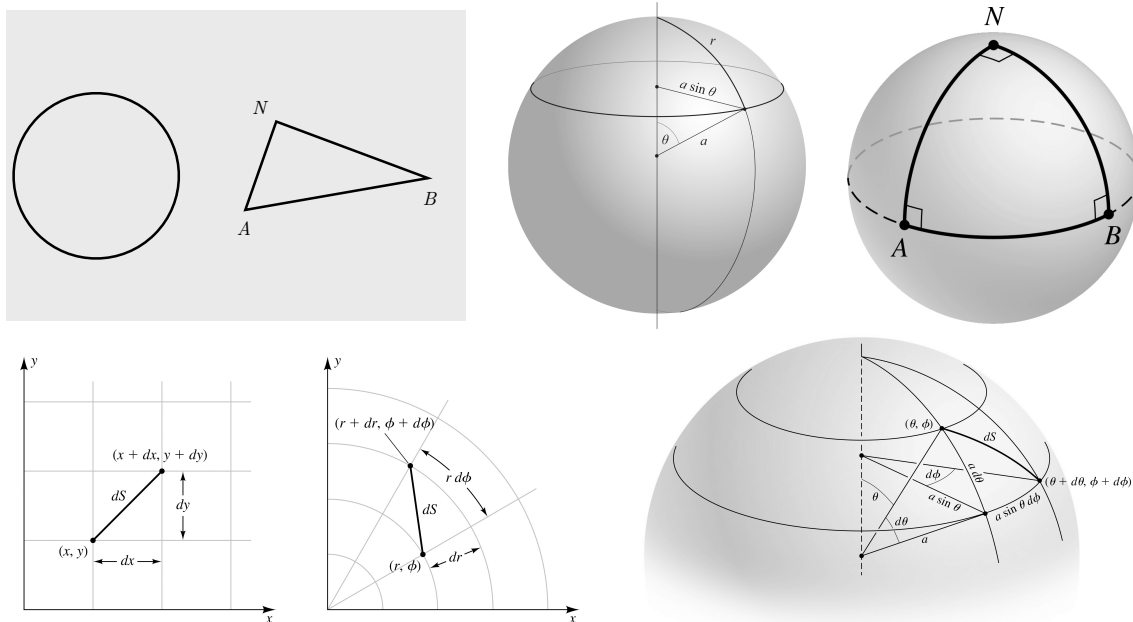


Figure 2.1: Top panel (right): The sum of the interior angles of a triangle ABN on a 2-sphere is $> \pi$, in this particular case it is $3\pi/2$. On the plane it is of course π . Top panel (middle sphere): The circumference of the circle of radius r is $< 2\pi r$, it is $C = 2\pi r \sin(r/a)/(r/a)$. On a plane it would be $C = 2\pi r$. Bottom panel (left): Cartesian and Polar co-ordinates in a 2d plane. Bottom panel (right): r and θ co-ordinates on the sphere. Part of the Fig. adapted from Hartle.

2.2 Specifying a geometry

How does one specify a geometry? A convenient and general way to do this is to define a rule for determining distances between nearby points. This is called a metric. Once this rule is specified, lengths of curves can be calculated by adding up such small distances. “Straight” lines can now be defined as curves with extremal lengths between two points. Circles (or spherical surfaces) can be defined as the locus of all points equidistant from a given point, angles can be defined as ratios of some arclength of a circle to the radius, and so on. This way of defining a geometry has significant advantages over trying to describe a geometry by embedding it in some higher dimensional Euclidean space or specifying it with a set of axioms. Both of these get cumbersome and difficult as we move to general geometries which do not have high degree of symmetry.

If the distances along the surface between all nearby cities of the earth were given, you would have been able to tell that the earth is not a plane. Indeed, if someone told you all the distances between any 4 four non-collinear cities, you would know that the surface of the earth is not planar. Please inform your Flat-Earther buddies.

2.2.1 Labeling and calculating with co-ordinates

Co-ordinates are labels of points in a geometry. For example, in a 2d plane, one can label points using the usual Cartesian co-ordinates (x, y) . One could also label points using polar co-ordinates

(r, θ) . Different co-ordinates can be chosen to label points in the same geometry. Different co-ordinate systems are connected by a co-ordinate transformation. For example, $x = r \cos \theta$ and $y = r \sin \theta$.

2.3 Spatial metric

The square of the distance, dS^2 , between nearby points in a plane with labels (x, y) and $(x + dx, y + dy)$ or (r, θ) and $(r + dr, \theta + d\theta)$

$$dS^2 = dx^2 + dy^2 = dr^2 + r^2 d\theta^2. \quad (2.3.1)$$

In different co-ordinate systems the expression for the distance can look different, as can be seen above. Here is the metric in yet another co-ordinate system, $dS^2 = 2d\tilde{x}^2 + d\tilde{y}^2 - 2d\tilde{x}d\tilde{y}$. This one also represents a 2d plane. The co-ordinates are linked by $(\tilde{x}, \tilde{y}) = (x, x + y)$. Changing co-ordinate systems does not change the geometry.

Exercise 2.3.1 Using $x = r \cos \theta$ and $y = r \sin \theta$, show that $dx^2 + dy^2 = dr^2 + r^2 d\theta^2$.

For a 3d Euclidean space, the distance between nearby points (x, y, z) and $(x + dx, y + dy, z + dz)$ or equivalently, (r, θ, ϕ) and $(r + dr, \theta + d\theta, \phi + d\phi)$ is

$$dS^2 = dx^2 + dy^2 + dz^2 = dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2). \quad (2.3.2)$$

The different co-ordinate systems are connected by $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$ and $z = r \cos \theta$.

Similarly, for a non-Euclidean geometry like a 2-sphere of radius a , a convenient choice of co-ordinates are the latitudes and longitudes (θ, ϕ) . The distance between nearby points (θ, ϕ) and $\theta + d\theta, \phi + d\phi$ is

$$dS^2 = a^2(d\theta^2 + \sin^2 \theta d\phi^2). \quad (2.3.3)$$

For a 3-sphere of radius a , the distance between nearby points (ψ, θ, ϕ) and $(\psi + d\psi, \theta + d\theta, \phi + d\phi)$ is

$$dS^2 = a^2[d\psi^2 + \sin^2 \psi(d\theta^2 + \sin^2 \theta d\phi^2)]. \quad (2.3.4)$$

While not necessary, you can embed this 3-sphere in 4d Euclidean space as a 3-surface $x^2 + y^2 + z^2 + w^2 = a^2$.

Note that this square of small distances can be conveniently written as

$$dS^2 = \sum_{i=1}^3 \sum_{j=1}^3 g_{ij} dx^i dx^j, \quad (2.3.5)$$

where $x^1 = \psi$, $x^2 = \theta$ and $x^3 = \phi$, where $g_{11} = a^2$, $g_{22} = a^2 \sin^2 \psi$, $g_{33} = a^2 \sin^2 \psi \sin^2 \theta$ with all other $g_{ij} = 0$ (the off-diagonal terms need not always be zero). We will use a simplified notation

$$dS^2 = g_{ij} dx^i dx^j, \quad (2.3.6)$$

where summation over repeated upstairs and down stairs indices is implied.⁴ Note that g_{ij} is a function of the co-ordinates $g_{ij}(x^1, x^2, x^3)$, which we will write as $g_{ij}(x^k)$ in shorthand.

For a given co-ordinate system, you can think of g_{ij} as components of a $d \times d$ matrix $\mathbf{g}$ where d is the dimension of space, with g_{ij} being the entry on the i -th row and j -th column. These entries will depend on the co-ordinate system. For example, in the case of the metric in 3d Euclidean space, the matrix entries in Cartesian co-ordinates would be $g_{ij} = \delta_{ij}$ where $\delta_{ij} = 1$ for $i = j$ and 0 for $i \neq j$. In Polar co-ordinates, $g_{11} = 1$, $g_{22} = r^2$ and $g_{33} = r^2 \sin^2 \theta$, and all other entries being 0. The components (matrix entries) look different, but together with $dx^i dx^j$ in the respective co-ordinates, they must yield the same dS^2 .

In general, the infinitesimal, $dS^2 = g_{ij} dx^i dx^j$, that determines the distance between any two nearby points is called the metric. In the case, since we are restricting ourselves to space and not spacetime, it is the spatial metric. It is also common practice to call g_{ij} the metric also.

Exercise 2.3.2 (i) Write down the components g_{ij} on a 2-sphere of radius a in terms of θ and ϕ co-ordinates. (ii) Consider the “Pinched Peanut” geometry defined by $dS^2 = a^2(d\theta^2 + f^2(\theta)d\phi^2)$ where $f(\theta) = \sin \theta(1 - \sin^2 \theta)$. Write down the components g_{ij} in terms of θ and ϕ co-ordinates.

2.3.1 Path length and volumes

Path length : To calculate non-infinitesimal distances along some curve γ , we can use the metric (in any co-ordinate system) as follows:

$$S[\gamma] = \int_{\gamma} dS = \int_{\lambda_1}^{\lambda_2} d\lambda \sqrt{g_{ij} \frac{dx^i}{d\lambda} \frac{dx^j}{d\lambda}}. \quad (2.3.7)$$

In the final expression, you should think of a parametrized curve $x^i(\lambda)$, and $g_{ij}(x^k(\lambda))$. The length of the curve is of course independent of the choice of co-ordinates, so it can be computed in any co-ordinate system.

Suppose Bob the bug is walking on a 2-sphere, leaving a localized chemical trail. Will the length of this chemical trail depend on the co-ordinates or parametrization we use to characterize the curve. Of course not! Nevertheless, depending on the geometry of space and the curve in question, it is useful to make a good co-ordinate choice for computational ease. It might be also useful sometimes to use one of the appropriately chosen co-ordinates itself as the parameter λ .

Important: Geometric quantities, for example, the distance between two nearby points, or the length of a particular curve, does not depend on the choice of co-ordinate systems or parametrization.

Exercise 2.3.3 Consider a 2d plane. A parabola passing through the origin can be parametrized as $x^1(\lambda) = \lambda^2/2$ and $x^2(\lambda) = \lambda$, where x^1 and x^2 are Cartesian co-ordinates. Write down g_{ij} and

⁴Notice the potentially confusing notation, $dx^2 \neq (dx)^2$ in these later expressions, although earlier we were using $dx^2 = (dx)^2$ earlier.

$dx^i(\lambda)/d\lambda$. Compute the length between $\lambda = 0$ and $\lambda = 1$. Hint: You might find the following substitution $\lambda = \sinh u$ useful for doing the integration.

Volumes To calculate the volume of a spatial region enclosed by a surface $\mathcal{S}$

$$\mathcal{V} = \int_{\mathcal{V}} \sqrt{\det g} \prod_{i=1}^d dx^i, \quad (2.3.8)$$

where $\det g$ is the determinant of g . To get some intuition, consider a metric which is diagonal in some co-ordinate system ($g_{ij} = 0$ for $i \neq j$). Then the infinitesimal distance along each co-ordinate direction when the co-ordinates change by dx^i is $dS_i = \sqrt{g_{ii}}dx^i$ (no summation). And the volume will be $\prod_{i=1}^d dS_i$. Note that the diagonality of g_{ij} means the co-ordinate curve intersect at right angles.

Exercise 2.3.4 Use the metric on the surface of a 2-sphere to do the following: (i) Calculate the circumference a circle of radius r (distance from the center to the circumference) in the two-dimensional geometry given by the surface of a sphere of radius a . Show that this reduces to $2\pi r$ when $r \ll a$. (ii) Calculate the area of the same circle on the sphere, and again show that it reduces to πr^2 when $r \ll a$. Bob, the bug lives here. (iii) Do the same for a cylinder of radius a and infinite extent in the perpendicular direction to the radius. Assume $r < \pi a$. Note that the cylinder is not globally Euclidean, but it is intrinsically flat.

Exercise 2.3.5 Beetle Bina lives in a different two-dimensional geometry with metric

$$dS^2 = dr^2 + a^2 \sinh^2(r/a) d\phi^2. \quad (2.3.9)$$

Here $0 \leq \phi < 2\pi$, and r is the intrinsic radial distance from $r = 0$. (i) Calculate the circumference $C(r)$ of a circle of radius r . (ii) Calculate the area $A(r)$ enclosed by this circle. (iii) Show that for $r \ll a$,

$$C(r) = 2\pi r \left[1 + \frac{r^2}{6a^2} + \cdots \right], \quad (2.3.10)$$

and

$$A(r) = \pi r^2 \left[1 + \frac{r^2}{12a^2} + \cdots \right]. \quad (2.3.11)$$

Compare these results with those for a Euclidean plane. For a circle of fixed radius r , does Bina have more area enclosed or does Bob? (See the previous problem for Bob).

Having hopefully gained some intuition about what intrinsically curved means, let me re-iterate that while we started with just curved space for pedagogy, ultimately we care about curved spacetime when thinking about gravitational physics.

For spacetime, the role played by Euclidean spatial geometry in the examples in this chapter, will be taken over by Minkowski spacetime geometry (which is non-Euclidean!).⁵ And, we will define intrinsically curved spacetime geometry by their departure from Minkowski spacetime geometry.

2.4 Practice Problems and Beyond

1. **Triangles on a Sphere:** Draw the following triangles on a 2-sphere (each side is part of a great circle). (i) The sum of interior angles is almost π . (ii) The sum of interior angles is almost 2π .
2. **Arc Lengths:** Using $dS = \sqrt{dx^2 + dy^2}$ show that the perimeter enclosing a half circle (the arc + straight line) of radius a , $C = \oint dS = \oint \sqrt{dx^2 + dy^2} = \pi a + 2a$. Similarly, use $dS = \sqrt{dr^2 + r^2 d\theta^2}$ in polar co-ordinates to find C also.
3. **More Peanuts:** Consider the peanut geometry described by the metric $dS^2 = a^2(d\theta^2 + f^2(\theta)d\phi^2)$ where $f(\theta) = \sin\theta(1 - 3\sin^2\theta/4)$. For $\theta = \Theta < \pi/2$ in this geometry, calculate the radius of the cap (you can't leave the surface again) enclosed by the circle defined by $\theta = \Theta$, and how the circumference of this circle is connected to the radius.
4. **Spherical Geometry 2d** Consider a latitude where $\theta = \Theta < \pi/2$ on a 2-sphere of radius a . Show that the radius of the spherical cap (you can't leave the surface!) enclosed by this latitude is $r = a\Theta$, the length of the latitude is $2\pi a \sin\Theta$. A useful generalization, if you want to know the distance between any two points on such a sphere (without restricting to the same latitude or longitude is as follows: The distance between two points on a 2-sphere or radius a , with co-ordinates (θ_1, ϕ_1) and (θ_2, ϕ_2) is

$$S = a \cos^{-1} [\sin\theta_1 \sin\theta_2 + \cos\theta_1 \cos\theta_2 \cos(\phi_1 - \phi_2)]. \quad (2.4.1)$$

See, for example, [here](#) for details.

5. **Not-so-flat Earth:** If the distances along the surface between all nearby cities of the earth were given, you would have been able to tell that the earth is not a plane. Indeed, if someone told you all the distances between any 4 four non-collinear cities, you would know that the surface of the earth is not planar. Let P_1, P_2, P_3 and P_4 be the four cities, with $S_{ij} = S_{ji}$ being the distance between them (length of the shortest curve in the surface between those points). Then, the 4-points lie in a plane if and only if:

$$\det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & S_{12}^2 & S_{13}^2 & S_{14}^2 \\ 1 & S_{21}^2 & 0 & S_{23}^2 & S_{24}^2 \\ 1 & S_{31}^2 & S_{32}^2 & 0 & S_{34}^2 \\ 1 & S_{41}^2 & S_{42}^2 & S_{43}^2 & 0 \end{pmatrix} = 0. \quad (2.4.2)$$

⁵The metric (in spatially Cartesian co-ordinates) is

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2. \quad (2.3.12)$$

It is the negative sign that makes it non-Euclidean.

Check this, and then please inform your flat-earther buddies. If you want to confirm that we are on an (idealized) 2-sphere, then the following must be true:

$$\det \begin{pmatrix} 1 & \cos S_{12} & \cos S_{13} & \cos S_{14} \\ \cos S_{21} & 1 & \cos S_{23} & \cos S_{24} \\ \cos S_{31} & \cos S_{32} & 1 & \cos S_{34} \\ \cos S_{41} & \cos S_{42} & \cos S_{43} & 1 \end{pmatrix} = 0. \quad (2.4.3)$$

For proofs, and more, see this post by Terence Tao [here](#).

2.5 Appendix

2.5.1 Curves of Extremal Length

The expression for $S[\gamma]$ is the length of any given curve γ :

$$S[\gamma] = \int_{\gamma} dS = \int_{\lambda_1}^{\lambda_2} d\lambda \sqrt{g_{ij} \frac{dx^i}{d\lambda} \frac{dx^j}{d\lambda}}. \quad (2.5.1)$$

Question: Given some points P and Q in our geometry, I want to find the curve connecting those points such that its length is a minimum (actually just “stationary”). The answer is provided by the curve $x^i(\lambda)$ which satisfies the **Euler-Lagrange equations**:

$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial (dx^i/d\lambda)} \right) - \frac{\partial L}{\partial x^i} = 0, \quad \text{where} \quad L = \sqrt{g_{ij} \frac{dx^i}{d\lambda} \frac{dx^j}{d\lambda}}. \quad (2.5.2)$$

Curves satisfying this equation have the property that between two fixed end points, if I try to change the curve γ to $\gamma + \delta\gamma$, the length of the curve does not change at first order in the small changes $\delta S[\gamma] = S[\gamma + \delta\gamma] - S[\gamma] = 0 + \mathcal{O}[\delta\gamma^2]$. The analogues of straight lines in arbitrary spatial geometries are called *geodesics*. They are curves whose length is stationary under sufficiently small variations with fixed endpoints, the ones we just found above.

Straight Line in a plane: Let me show you the use of this “technology” to go through a long route to find the equation of a straight line in a 2d plane. While seemingly trivial, understanding how it is used here will help you gain confidence in using it in more complicated situations later. I will also introduce the (sometimes confusing) notion of an affine parameter.

Consider two points P and Q in a plane. Let us put Cartesian co-ordinates (x, y) on the plane. Consider a collection of curves with P and Q as their endpoints. Each curve is parametrized by λ : $(x(\lambda), y(\lambda))$. We move along a curve from P to Q as λ changes. We wish to find the curve connecting P and Q with extremal length. Of course you intuitively know that this must be a straight line, ie. $dy/dx = \text{const}$. We will prove this. In terms of the parameterized curve, you perhaps might also be expecting the equation that the curve satisfies is $d^2x(\lambda)/d\lambda^2 = 0$ and $d^2y(\lambda)/d\lambda^2 = 0$. We will see that this will not be true in

general for arbitrary choice of λ , but will turn out to be true a special class of them. This special class is: $\lambda = aS + b$ where S is the arc length along the curve for arbitrary choice of $a \neq 0$ and b .

Let us begin. The length of any curve γ from P to Q

$$\begin{aligned} S[\gamma] &= \int_{\gamma} dS = \int \sqrt{dx^2 + dy^2} = \int d\lambda \sqrt{\left(\frac{dx}{d\lambda}\right)^2 + \left(\frac{dy}{d\lambda}\right)^2}, \\ &= \int d\lambda L, \quad \text{where} \quad L = \sqrt{\left(\frac{dx}{d\lambda}\right)^2 + \left(\frac{dy}{d\lambda}\right)^2}. \end{aligned} \quad (2.5.3)$$

If a curve extremizes its length, then $\delta S = 0$. Such a curve satisfies the Euler-Lagrange equations

$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial(dx/d\lambda)} \right) = \frac{\partial L}{\partial x}, \quad \frac{d}{d\lambda} \left(\frac{\partial L}{\partial(dy/d\lambda)} \right) = \frac{\partial L}{\partial y}. \quad (2.5.4)$$

These equations yield:

$$\frac{d}{d\lambda} \left(\frac{1}{L} \frac{dx}{d\lambda} \right) = 0, \quad \frac{d}{d\lambda} \left(\frac{1}{L} \frac{dy}{d\lambda} \right) = 0. \quad (2.5.5)$$

Since these equations imply $L^{-1}dx/d\lambda = \text{const.}$ and $L^{-1}dy/d\lambda = \text{const.}$, taking the ratio of these constants is also a constant. That is $dy/dx = \text{const.}$ This is of course the expected equation for the locus of a line.

However, these equations do not yield $d^2x/d\lambda^2 = 0$ and $d^2y/d\lambda^2 = 0$. Now, look back at the equation for S . You will notice that $d\lambda L = dS$, where dS is the infinitesimal length along the curve. So our equation for $x(\lambda)$ can be written as (after multiplying by L^{-1})

$$\frac{1}{L} \frac{d}{d\lambda} \left(\frac{1}{L} \frac{dx}{d\lambda} \right) = 0 \implies \frac{d^2x}{dS^2} = 0, \quad (2.5.6)$$

where we used $Ld\lambda = dS$. Similarly for the y equation.

Are there a general class of parameters λ' for which $d^2x/d\lambda'^2 = 0$? Clearly if $dS = d\lambda'/a$ or equivalently, if $\lambda' = aS + b$, then again $d^2x/d\lambda'^2 = 0$. Similarly for y . Thus all parameters λ' which are related linearly to the distance along the curve S will yield the desired simple equations. These are called **affine parameters**.

Note that regardless of the parametrization, the locus is $dy/dx = \text{const.}$ – it is a line as you expect. However $d^2x/d\lambda^2 = 0$ and $d^2y/d\lambda^2 = 0$ for along this straight line locus of points only if we chose λ to be an affine parameter.

FROM NEWTONIAN SPACE AND TIME TO MINKOWSKI SPACETIME

I am combining aspects of Ch. 3 and Ch. 4 of Hartle here. You should read the chapters there first, this is still in progress here, will be done after the lectures.

3.1 Newtonian Space, Time and Reference Frames

The arena of Newtonian physics is a 3-dimensional Euclidean space and a universal time.

Reference Frames In Euclidean space, imagine an idealized 3d rigid lattice of rods. At each intersection there is an idealized clock, all ticking at a universal rate. I like to think of tiny demons sitting at these intersections, meticulously recording whatever happens in their vicinity. This lattice of rods and clocks defines a *reference frame*. It allows us to label where and when events take place.

Although the lattice naturally suggests Cartesian co-ordinates, we could equally well use spherical polar co-ordinates, or some other choice. How we label points is a choice; the reference frame is the same. Furthermore, not just co-ordinates of a given reference frame, but the choice of reference frame itself does not change the underlying (assumed Euclidean) geometry of space, and universality of time.

Different reference frames can be displaced, rotated, rotating, or moving with respect to each other. Among this large family, there is a special class of frame called *inertial frames*. Within the Newtonian spacetime context, we define them below.

3.1.1 Inertial Reference Frames

Go far away from large masses and shield yourself, as well as you can, from other long-range forces. Set up a reference frame with Cartesian co-ordinates (x, y, z) and synchronized clocks. If a free particle moving through the frame satisfies

$$\frac{d^2x}{dt^2} = \frac{d^2y}{dt^2} = \frac{d^2z}{dt^2} = 0, \quad (3.1.1)$$

then your frame is *inertial*. Any frame moving with constant velocity with respect to an inertial frame is also inertial. A rotated (but not rotating) or displaced version of an inertial frame is also inertial.

3.2 Galilean Transformations

Consider two inertial frames $\mathcal{S}$ and $\mathcal{S}'$ in the standard configuration: their axes are aligned, their origins coincide at $t = t' = 0$, and $\mathcal{S}'$ moves with speed v along the x -axis of $\mathcal{S}$. Then, the relationships between the co-ordinates and time of an event is give by

$$x = x' + vt', \quad y = y', \quad z = z', \quad t = t'. \quad (3.2.1)$$

For a particle with velocity $(V^{x'}, V^{y'}, V^{z'})$ in $\mathcal{S}'$,

$$(V^x, V^y, V^z) = (V^{x'} + v, V^{y'}, V^{z'}). \quad (3.2.2)$$

3.3 Principle of Relativity

No experiment performed entirely inside an inertial laboratory can distinguish one inertial frame from another.

Consider two inertial frames, which we will imagine to be enclosed opaque boxes (with grids and clocks inside them as before), with our previous $\mathcal{S}$ being Sammy's frame and $\mathcal{S}'$ being Polly's. The Principle of Relativity states that if Sammy and Polly set up exactly the same initial configurations of matter and fields in their respective inertial laboratories, the subsequent evolution of the matter and fields will be identical in their respective frames. That is, the laws of physics are the same in all inertial frames.

Galileo, said it best (from his *Dialogue Concerning the Two Chief World Systems*):

Shut yourself up with some friend in the main cabin below decks on some large ship, and have with you there some flies, butterflies, and other small flying animals. Have a large bowl of water with some fish in it; hang up a bottle that empties drop by drop into a wide vessel beneath it. With the ship standing still, observe carefully how the little animals fly with equal speed to all sides of the cabin. The fish swim indifferently in all directions; the drops fall into the vessel beneath; [...] When you have observed all these things carefully [...], have the ship proceed with any speed you like, so long as the motion is uniform and not fluctuating this way and that. You will discover not the least change in all the effects named, nor could you tell from any of them whether the ship was moving or standing still. In throwing something to your companion, you will need no more force to get it to him whether he is in the direction of the bow or the stern, with yourself situated opposite. [...] the butterflies and flies will continue their flights indifferently toward every side, nor will it ever happen that they are concentrated toward the stern, as if tired out from keeping up with the course of the ship, from which they will have been separated during long intervals by keeping themselves in the air.

Note that Galileo was talking about mechanical phenomenon, our principle, as used by Einstein applies to all phenomenon.

3.3.1 “Light” Trouble for Galilean Transformations

If Sammy and Polly set up identical experiments with a laser in their respective laboratories and measure the speed of light, they both find $c = 3 \times 10^{10} \text{ cm s}^{-1}$. This is consistent with the Principle of Relativity (PoR).

Now let them drill holes in their laboratories so that light emitted by Polly’s laser enters Sammy’s laboratory. Sammy measures the speed of this incoming light and again finds it to be c .¹ Note that this is a different statement from the previous one: before, Sammy and Polly performed identical experiments within their own laboratories; now they are describing the same physical light sent from Polly’s laboratory to Sammy’s.

This spells trouble for Galilean Transformations (GT). Suppose Polly’s reference frame moves with velocity $v\hat{x}$ relative to Sammy’s, and Polly sends the light toward Sammy in the $-\hat{x}$ direction. Polly measures the velocity of the light to be $-c\hat{x}$. According to GT, Sammy should therefore measure its velocity to be $(v - c)\hat{x}$, and hence its speed should be $|v - c|$. But Sammy instead measures the speed of the light to be exactly $c = 3 \times 10^{10} \text{ cm s}^{-1}$, independent of the relative velocity v between the two laboratories.

The prediction $|v - c|$ arose directly from GT. If we retain the Principle of Relativity and the observed behavior of light, GT must therefore be wrong.

3.3.2 “Light” Trouble for Galilean Transformations

If Sammy and Polly set up identical experiments with a laser in their respective laboratories and measure the speed of light, they both find it to be the same speed $c = 3 \times 10^{10} \text{ cm s}^{-1}$. This is consistent with the Principle of Relativity (PoR).

Now, let them both drill a hole in their laboratories in such a way that Polly’s laser light enters Sammy’s laboratory. Sammy measures this speed, and also finds it to be c . Note that this is different from the previous result where c was measured under identical setups within each laboratory.

This spells trouble for Galilean Transformations (GT). For according to GT, the speed of this light sent by Polly to Sammy, as measured by Sammy in his reference frame should be $(v - c)\hat{x}$ where c is the speed of light measure by Polly in her frame ($= 3 \times 10^{10} \text{ cm s}^{-1}$). Sammy measures this speed to be exactly $3 \times 10^{10} \text{ cm s}^{-1}$ also, independent of the speed v of Polly’s reference frame relative to Sammy’s. Recall that $v - c$ came about because of GT. That means, GT must be wrong!

3.4 Resolution via Lorentz Transformations

We can leave PoR as is, but change GT as follows. Einstein suggested the transformation between frames should be²

$$x = \gamma(x' + vt'), \quad y = y', \quad z = z', \quad t = \gamma(t' + vx'/c^2), \quad (3.4.1)$$

¹This is what Michelson-Morley essentially found, though they were really looking for motion relative to ether.

²They carry Lorentz name because he had already discovered these transformations in the context of “ether” theory. Einstein gave them a fundamentally different interpretation.

where

$$\gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}}. \quad (3.4.2)$$

With these transformations, the velocities are related by

$$(V^x, V^y, V^z) = \left(\frac{V^{x'} + v}{1 + vV^{x'}/c^2}, \frac{V^{y'}}{\gamma(1 + vV^{x'}/c^2)}, \frac{V^{z'}}{\gamma(1 + vV^{x'}/c^2)} \right). \quad (3.4.3)$$

When $v/c \ll 1$, this reduces to the Galilean result. But if $V^{x'} = \pm c$, then

$$V^x = \pm c, \quad (3.4.4)$$

independent of v .

3.5 The “Price”

We have retained the Principle of Relativity and the observed behavior of light, but replaced Galilean Transformations by Lorentz Transformations. There is a price: the Newtonian notions of a universal time and an observer-independent spatial distance can no longer be maintained.

Recall that under Galilean Transformations, $x = x' + vt'$, $t = t'$. In particular, if two events occur at the same time for Sammy, $dt = 0$, then they also occur at the same time for Polly, $dt' = 0$.

Thus, simultaneity is universal in Newtonian physics: if two spatially separated events happen “at the same time” for one inertial observer, they happen at the same time for every inertial observer.

Moreover, in Newtonian physics, the underlying geometry of space is Euclidean, and the distance between two nearby points in Cartesian co-ordinates *at a given instant* is $dS^2 = dx^2 + dy^2 + dz^2$. All inertial observers agree on this distance.

Now? : Recall that with Lorentz transformations $x = \gamma(x' + vt')$ and $t = \gamma(t' + vx'/c^2)$. For two nearby events,

$$dt = \gamma \left(dt' + \frac{v}{c^2} dx' \right). \quad (3.5.1)$$

Therefore, if Sammy regards the two events as simultaneous, $dt = 0$, then Polly finds

$$dt' = -\frac{v}{c^2} dx' \neq 0, \quad (3.5.2)$$

for spatially separated events ($dx' \neq 0$).

Two events that are simultaneous for Sammy need not be simultaneous for Polly. There is no universal notion of “now.”

Length? : This also changes our notion of spatial distance. Measuring the length of an extended object requires determining the positions of its two ends *simultaneously*, that is, at the same time. Since Sammy and Polly no longer agree on which pairs of events are simultaneous, they need not agree on the spatial length of an object. Spatial distances, like time intervals, can depend on the inertial frame.

3.6 Invariant spacetime interval and Minkowski spacetime

What replaces these separately observer-independent notions of space and time is an observer-independent quantity associated with *spacetime*. Define

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2. \quad (3.6.1)$$

Using Lorentz Transformations, one finds

$$ds^2 = -c^2 dt'^2 + dx'^2 + dy'^2 + dz'^2. \quad (3.6.2)$$

Thus, Sammy and Polly need not agree separately on spatial distances or time intervals, but they agree on the spacetime interval ds^2 .

Exercise 3.6.1 For the standard configuration of reference frames, show that $-c^2 dt^2 + dx^2 + dy^2 + dz^2 = -c^2 dt'^2 + dx'^2 + dy'^2 + dz'^2$.

The arena of physics is therefore no longer naturally separated into a universal 3-dimensional Euclidean space, and a universal time. Space and time must instead be regarded as parts of a single 3+1 dimensional spacetime.

The geometry of this spacetime is defined by the spacetime metric

$$ds^2 = -c^2 dt^2 + dS^2, \quad (3.6.3)$$

where dS^2 is the Euclidean metric. The geometry defined by ds^2 is the **Minkowski spacetime**. Notice the minus sign. Minkowski spacetime is not Euclidean. But it is flat.

As before, the underlying geometry is independent of our choice of frame or co-ordinates. We are free to use spherical polar co-ordinates for space, in which case

$$ds^2 = -c^2 dt^2 + dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (3.6.4)$$

The expression looks different, but the spacetime geometry is the same.

Reference frames in Minkowski spacetime We can again imagine a lattice of rods and clocks. The clocks now need to be synchronized carefully. A convenient operational prescription is Einstein synchronization: if one clock sends a light signal at t_e , a nearby clock reflects it, and the first receives it at t_r , then the nearby clock is assigned the reflection time $t_{\text{bounce}} = \frac{1}{2}(t_e + t_r)$.

3.7 Moving in Spacetime

The best way to explore Minkowski spacetime is to think about the paths of particles through it.

For simplicity, consider 1 + 1 dimensions. We draw ct and x on an ordinary sheet of paper, but remember that the geometry of the diagram is not Euclidean. Two distinct events can even satisfy $ds^2 = 0$, something impossible for Euclidean distance.

For two nearby events:

$$ds^2 < 0 \quad \text{timelike,} \quad (3.7.1)$$

$$ds^2 = 0 \quad \text{lightlike or null,} \quad (3.7.2)$$

$$ds^2 > 0 \quad \text{spacelike.} \quad (3.7.3)$$

The locus of particles (massive or massless) in spacetime are called world lines. Massive particles move on timelike worldlines – that is $ds^2 < 0$ along every segment of the worldline. On the other hand, light moves on null worldlines with $ds^2 = 0$.

3.8 Proper Time

For a timelike worldline we define the **proper time** by

$$d\tau^2 = -\frac{ds^2}{c^2}. \quad (3.8.1)$$

Thus

$$\boxed{d\tau = dt \sqrt{1 - \frac{1}{c^2} \left(\frac{d\mathbf{x}}{dt} \right)^2}} \quad (3.8.2)$$

If you are momentarily at rest with the particle, then $d\mathbf{x} = 0$, and

$$d\tau = dt. \quad (3.8.3)$$

So proper time is simply the time measured by a clock moving along the worldline. It is a geometric, frame-independent quantity.

For a worldline connecting events P and Q ,

$$\tau_{PQ} = \int_P^Q d\tau. \quad (3.8.4)$$

This is directly analogous to the length of a spatial curve that we discussed in the previous chapter.

A free massive particle follows a straight worldline in Minkowski spacetime. Equivalently, among nearby timelike curves connecting the same two events, its worldline makes the proper time stationary. In flat spacetime, the straight timelike path gives the maximum proper time.

Between two timelike separated events, the inertial straight-line worldline has maximal proper time.

This is the clean geometric core of the twin paradox, you can read more about this in Hartle.

3.9 Spacetime Diagrams

Spacetime diagrams are invaluable for keeping track of events and comparing different frames. Always label the events you are comparing.

For a frame $\mathcal{S}'$ moving at speed v along x relative to $\mathcal{S}$:

Time axis of $\mathcal{S}'$ The origin of $\mathcal{S}'$ satisfies $x' = 0$, so

$$x = vt. \quad (3.9.1)$$

Space axis of $\mathcal{S}'$ Events with $t' = 0$ satisfy

$$ct = \frac{v}{c}x. \quad (3.9.2)$$

These tilted axes already show that simultaneity is frame-dependent.

Time dilation For a clock at rest in $\mathcal{S}'$, $\Delta x' = 0$, so invariance of the interval gives

$$c^2 \Delta t'^2 = c^2 \Delta t^2 - \Delta x^2. \quad (3.9.3)$$

Since $\Delta x = v \Delta t$,

$$\boxed{\Delta t' = \frac{\Delta t}{\gamma}.} \quad (3.9.4)$$

A clock moving relative to us accumulates less proper time between the same two events.

Length contraction A rod of proper length L is measured in a frame in which it is moving by recording the positions of its endpoints at the same time in that frame. The result is

$$\boxed{L_{\text{moving}} = \frac{L}{\gamma}.} \quad (3.9.5)$$

Be careful: length measurements in two different frames generally involve different pairs of simultaneous events.

3.10 Newtonian Physics as the Low-Speed Limit

For $V \ll c$,

$$d\tau = dt \sqrt{1 - \frac{V^2}{c^2}} \simeq dt \left(1 - \frac{V^2}{2c^2} + \dots \right). \quad (3.10.1)$$

The relativistic free-particle action

$$S_{\text{free}} = -mc^2 \int d\tau \quad (3.10.2)$$

therefore becomes

$$S_{\text{free}} \simeq -mc^2 \int dt + \int dt \frac{mV^2}{2} + \dots. \quad (3.10.3)$$

Apart from the irrelevant constant term, the familiar Newtonian kinetic action is recovered.

Thus Newtonian physics is not discarded. It is the low-speed limit of the spacetime geometry of Special Relativity.

3.11 Take Away

Newtonian physics assumes a Euclidean 3-space and a universal time. Galilean transformations preserve this separation.

The constancy of the speed of light forces us to abandon universal time. Space and time mix under Lorentz transformations, but the spacetime interval

$$\boxed{ds^2 = -c^2 dt^2 + dS^2} \tag{3.11.1}$$

is invariant.

This is the geometry of Minkowski spacetime. Proper time is the length of a timelike worldline, and free particles move on straight worldlines.

The next step is to ask what happens when spacetime itself is curved.